SIMATS ENGINEERING

ASSESSMENT TOOL 2: Concept Mapping

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Assessment Tool Weightage: 20%

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 20**
Question Count: 4

Code of Conduct

I Manoj M (192511060) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

Assessment Tasks

Concept Mapping Tasks

Create concept maps for the following tasks. Each map must show key nodes, link labels, and a logical hierarchy.

1. Concept Map 1: Cloud Computing Fundamentals. Include definition, characteristics, benefits, and limitations.
2. Concept Map 2: Evolution of Cloud Computing. Connect hardware evolution, internet software evolution, distributed systems, and service delivery.
3. Concept Map 3: Service Models. Map IaaS, PaaS, SaaS, and XaaS with provider responsibilities and user responsibilities.
4. Concept Map 4: Virtualization and Web Services as Enablers of Cloud. Show how these two concepts support cloud deployment and scalability.

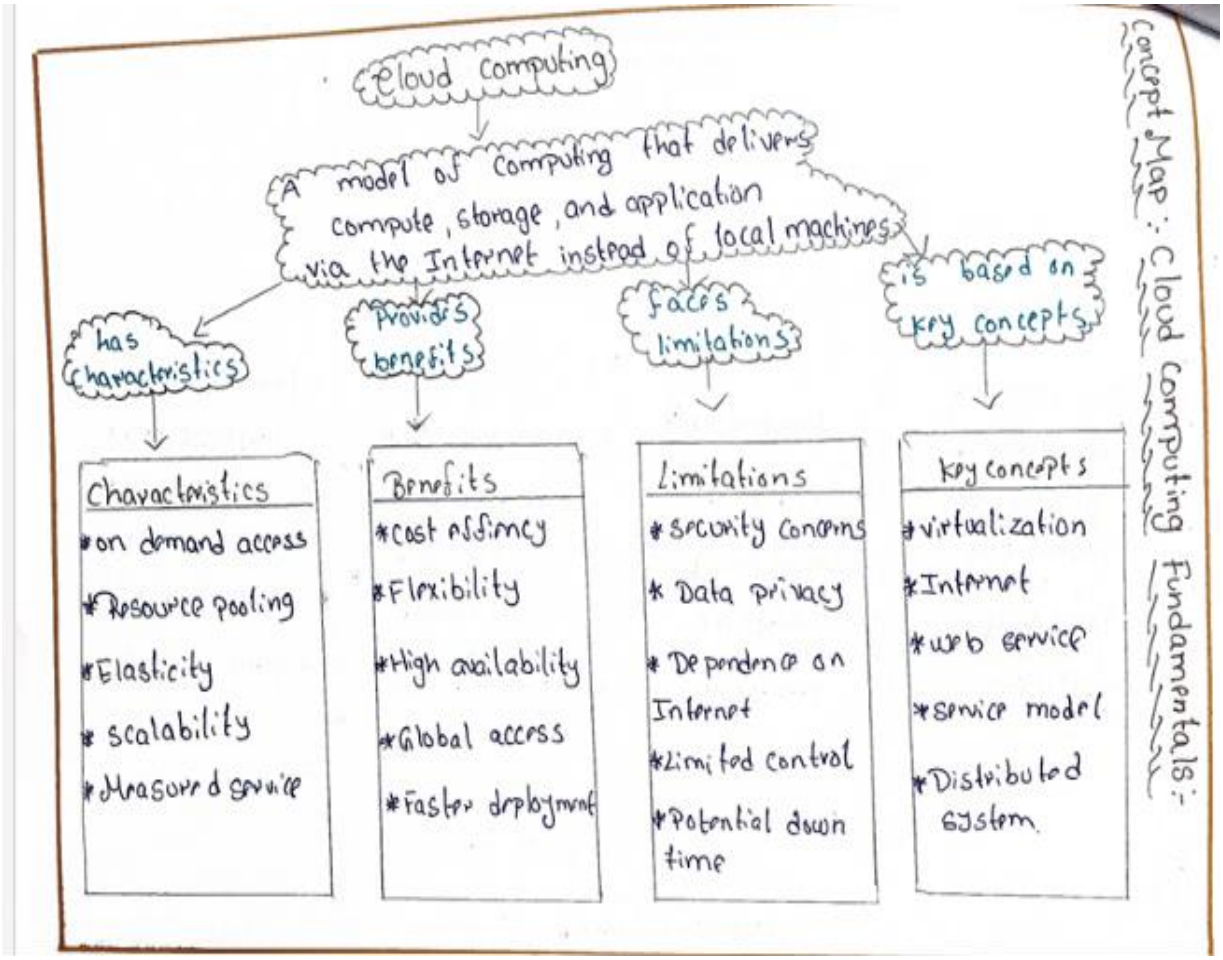
Expected concept nodes:

- Elasticity, scalability, resource pooling, on-demand access, measured service
- Virtual machine, hypervisor, abstraction, API, SOAP, REST, provider, consumer
- Infrastructure layer, platform layer, application layer, shared responsibility

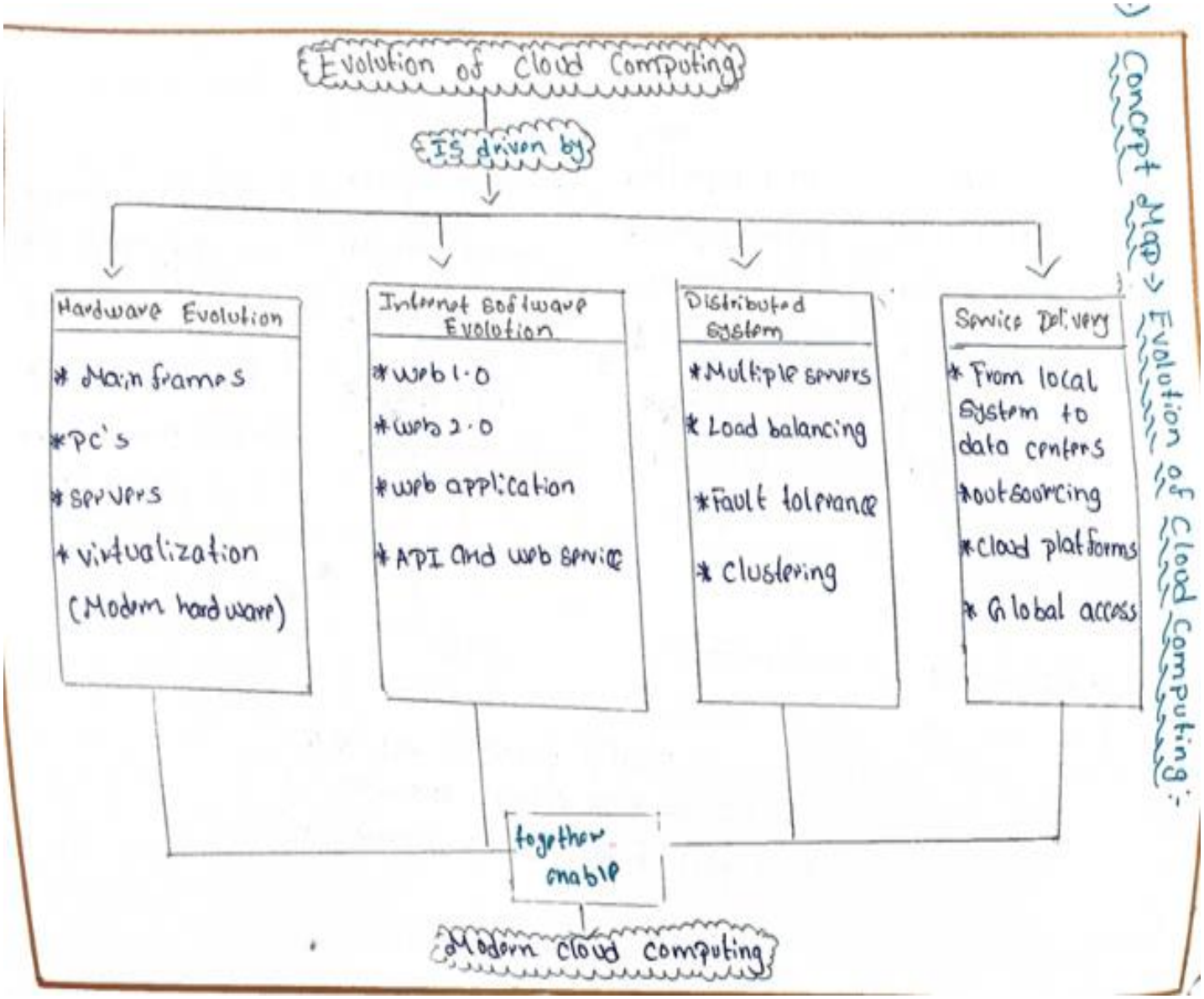
Concept Mapping Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Concept Identification	25%	All major concepts are accurately identified	Most concepts are identified correctly	Limited concepts are identified	Essential concepts are missing
Relationship	25%	Links between	Most links are	Some links are	Relationships are

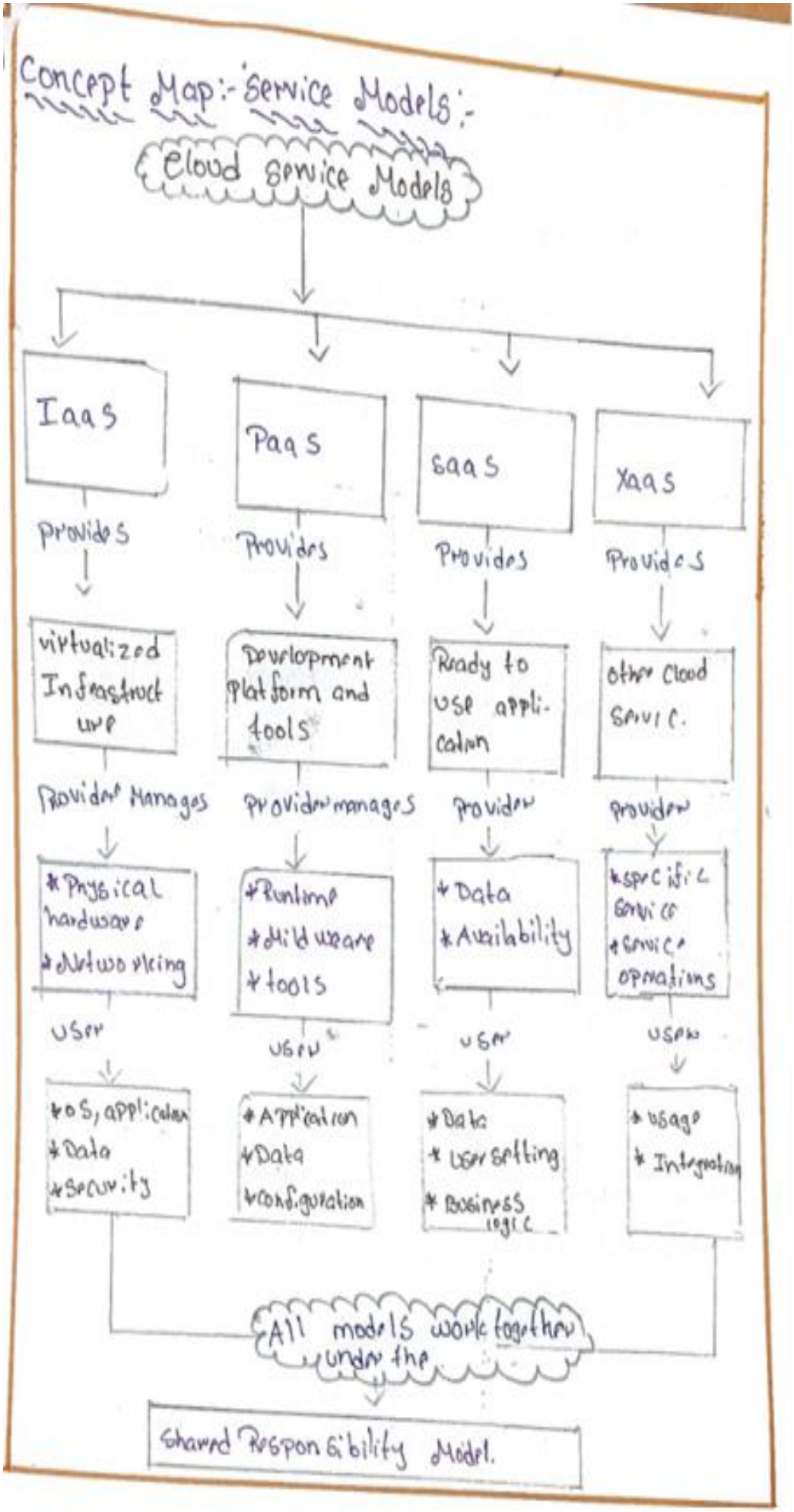
Mapping		concepts are accurate and meaningful	correct	weak or unclear	mostly incorrect
Hierarchy and Organization	20%	Excellent structure with clear hierarchy	Mostly organized with minor issues	Basic structure with weak hierarchy	No logical organization
Technical Relevance	20%	Map strongly reflects cloud course content	Mostly aligned to topic	Partially aligned to topic	Weak alignment to topic
Visual Clarity	10%	Neat, readable, and easy to interpret	Generally readable	Somewhat cluttered	Poorly presented


Concept Map : Cloud Computing Fundamentals :
Cloud computing
A model of computing that delivers compute, storage, and application via the Internet instead of local machines
has characteristics
Provides benefits
faces limitations
is based on key concepts
Characteristics
*on demand access
*Resource pooling
*Elasticity
* scalability
*Measured service
Benefits
*cost effiency
*Flexibility
*High availability
*Global access
*Faster deployment
Limitations
* security concerns
* Data privacy
* Dependence on Internet
*Limited control
*Potential down time
key concepts
*virtualization
*Internet
*web service
*service model
*Distributed system

2.


Concept Map → Evolution of Cloud Computing :
Evolution of Cloud Computing
Is driven by
Hardware Evolution
* Main frames
* PC's
* servers
* virtualization
(Modern hardware)
Internet Software Evolution
* web 1.0
* web 2.0
* web application
* API and web service
Distributed System
* Multiple servers
* Load balancing
* Fault tolerance
* Clustering
Service Delivery
* From local system to data centers
* outsourcing
* cloud platforms
* Global access
together enable
Modern cloud computing

Concept Map:- Service Models:-



4) virtualization and web service as enablers of cloud.

